

Coins and Conflict

Spatial Patterns in the Archaeological Record

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Research Question

Can spatial statistical methods recover coherent and interpretable patterns from incomplete archaeological evidence?

Do spatial statistical methods recover interpretable signal from the CHRR dataset, and does the pattern of recovery correspond to documented historical events?

Coin Hoard Data^{1,2}

Michael Crawford¹



Assembled a hoard and coin corpus

Standardized and developed a descriptive framework

Hoard¹

Name: <http://numismatics.org/chrr/id/ORI>

Closing Date: 74 BCE

Findspot: Oristà, Spain ± 5km



Coin



CHRR Database²



Resource Description Framework (RDF) accessed
by SPARQL query

Reference: RRC 210/1

URI: <http://numismatics.org/crrr/id/rrc-210.1>

Mint Date: 149 BCE

Denomination: Denarius

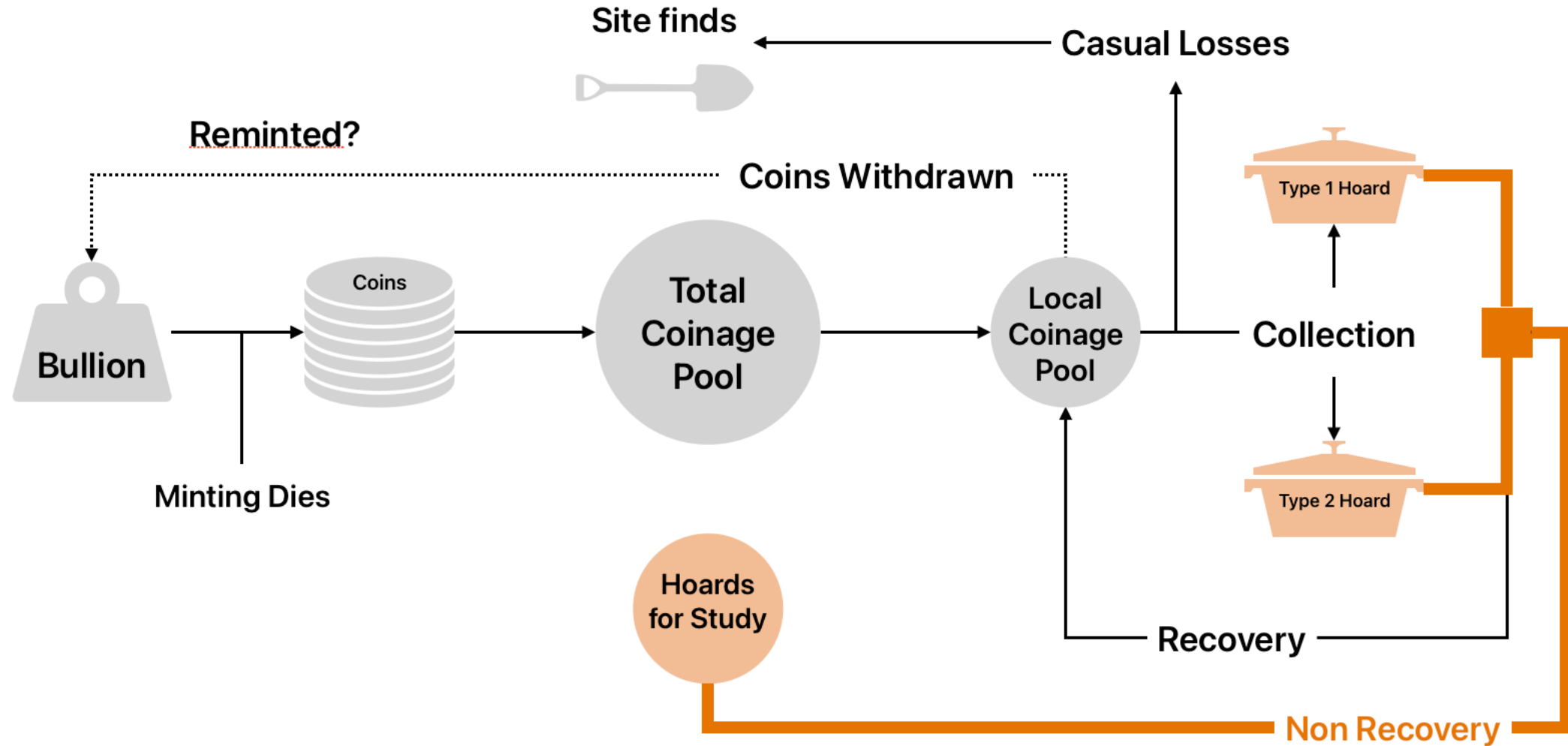
Mint: Rome



¹ Crawford (1969). *Roman Republican Coin Hoards*. Royal Numismatic Society.

² Lockyear, Kris (2013). *Coin hoards of the Roman Republic Online*, version 1.0. New York: American Numismatic Society. Data retrieved from <https://numismatics.org/chrr/> on 04/03/26

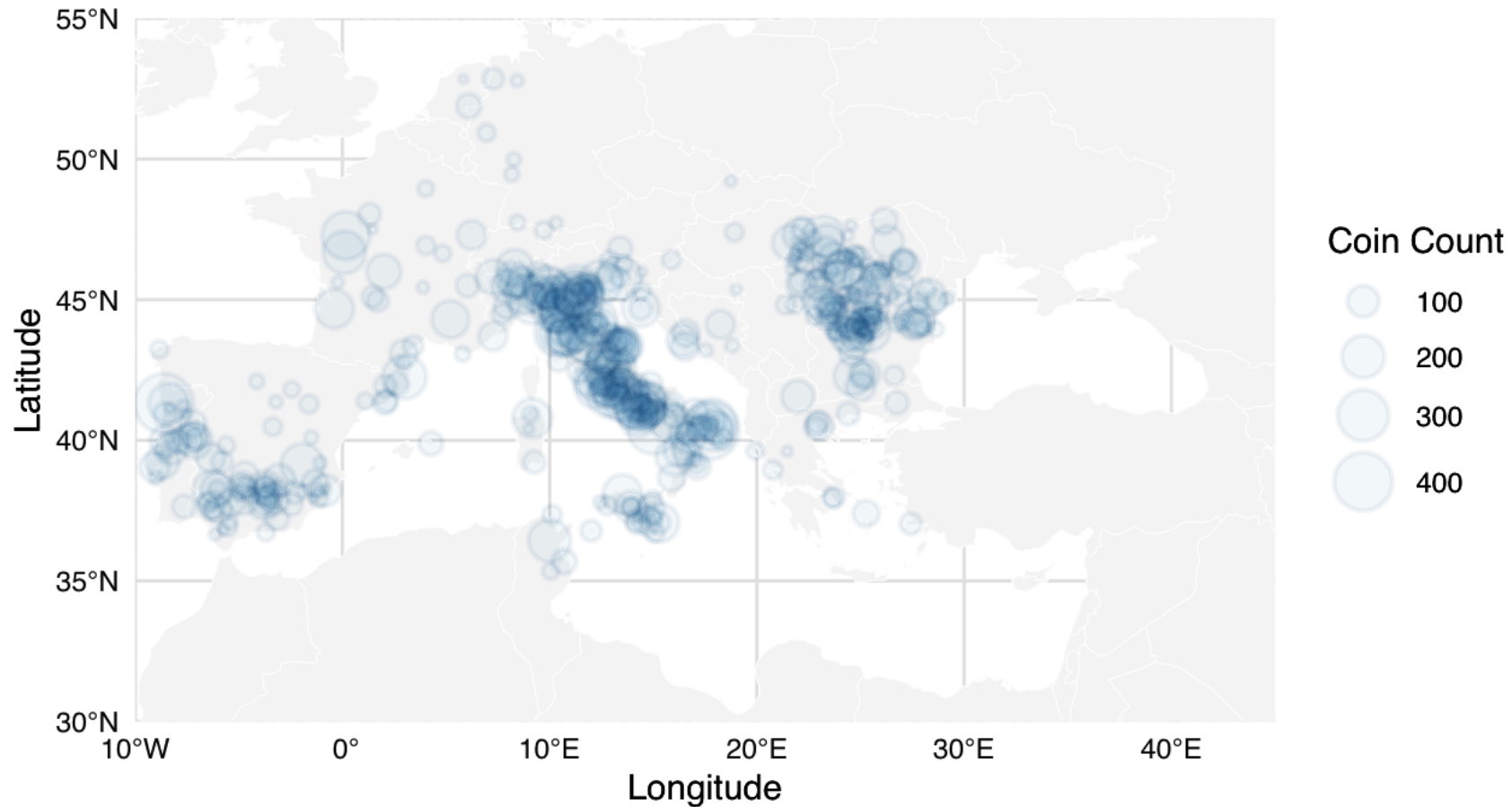
Coin Hoard Creation¹



¹ Lockyear, K. (1991). *Simulating coin hoard formation*. British Archaeological Reports.

Primary Dataset^{1,2}

Coin Hoards of the Roman Republic (CHRR)



Data: Coin Hoards of the Roman Republic (CHRR)

¹ Crawford (1974). *Roman Republican Coinage*. Cambridge University Press.

² Lockyear, Kris (2013). Coin hoards of the Roman Republic Online, version 1.0. New York: American Numismatic Society. Data retrieved from <https://numismatics.org/chrr/> on 04/03/26

Core Challenge - Recovering the Signal

Sampling Bias

Ergodicity - Observed data isn't a true sample. Hoards are survivors not samples.

System Instability

Spatial Non-Stationarity - Spatial relationships change over time.

Recoverability

Compounding Errors - Biases do not cancel, only accumulate.

Examples of Coin Analyses

Composition & Interpretation

Author	Analyses
Bland (2020)	Empire-wide hoard density comparison; collection bias framework
Buttrey (1999)	Critique: composition reflects deposition, not just circulation
Crawford (1969, 1974)	Systematic hoard catalogue and typological framework
Lockyear (1991, 2007)	Correspondence analysis, simulation, and seriation of hoard composition
Preston (1983)	Age correction for hoard counts; exponential decay model for coin attrition

Monetary Dynamics

Author	Analyses
Crawford (1974)	Die-count production estimates and chronological dating system
de Callatay (1995)	Defended quantification of mint output; new Republican die studies.
Howgego (1995)	Narrative reconstruction of circulation patterns and monetary supply
Kemmers (2019)	Review of coin function: military pay, commerce, and store of value
Lockyear (1991)	Simulated hoard formation under production and circulation assumptions

Geography and Mapping

Author	Analyses
Găzdac et al. (2007)	Findspot maps of Republican coins in Transylvania
Lockyear & Gruber (2013)	CHRR digitized dataset with hoard coordinates
Ollive et al. (2008)	Coin distribution mapped onto DEM to track settlement dynamics
Reece (1995)	Geographic distribution of site-finds across Roman Britain
Vicari Sottosanti et al. (2025)	GIS spatial analysis of Roman antoniniani in Sicily

Geospatial Analysis

Author	Analyses
Haddad & Araújo (2022)	Coin finds as spatial economic evidence
Foldvari et al. (2022)	Probit model of interregional coin diffusion velocity using mint-to-find-spot distance

Secondary Datasets

ORBIS¹: Stanford geospatial network model of the Roman world road and water travel time in days from Rome to each hoard location.

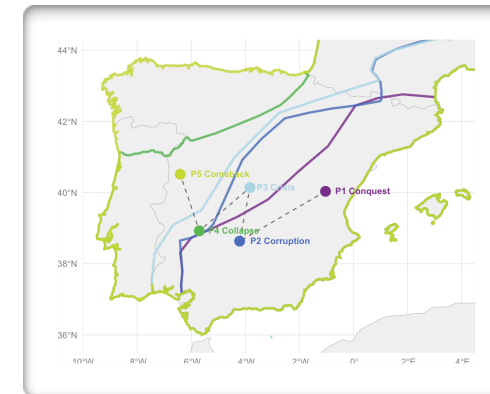


Custom Shapefiles²: digitized political boundary polygons matched to each study period, and a maximum extent boundary for Dacia

Custom Periods³: Time was divided up in to custom periods to reflect changes in the Roman Republic

Natural Earth³: Base maps and natural feature geometry

Pleiades⁴: Geospatial and other data ancient locations.



¹ Scheidel, W., Meeks, E., & Talbert, R. (2012). ORBIS: The Stanford Geospatial Network Model of the Roman World. Stanford University. <https://orbis.stanford.edu>

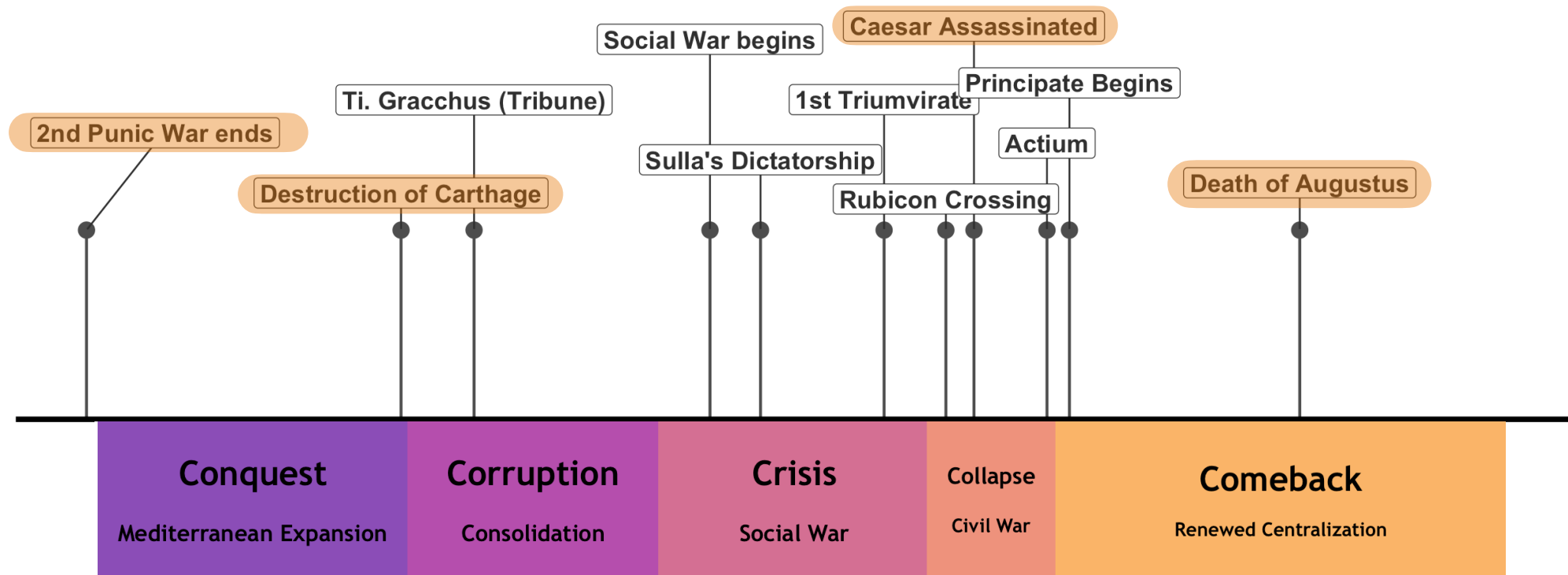
² David Hollander, Roman Empire Extent Shapefiles (2026), personal communication.

³ David Hollander, Roman Republican Periodization Scheme (2026), personal communication.

⁴ Natural Earth, *Free Vector and Raster Map Data*, <https://www.naturalearthdata.com>.

⁵ Elliott, T., et al. *Pleiades: A Gazetteer of Past Places*. New York University & Institute for the Study of the Ancient World. <https://pleiades.stoa.org>

Historical Periods



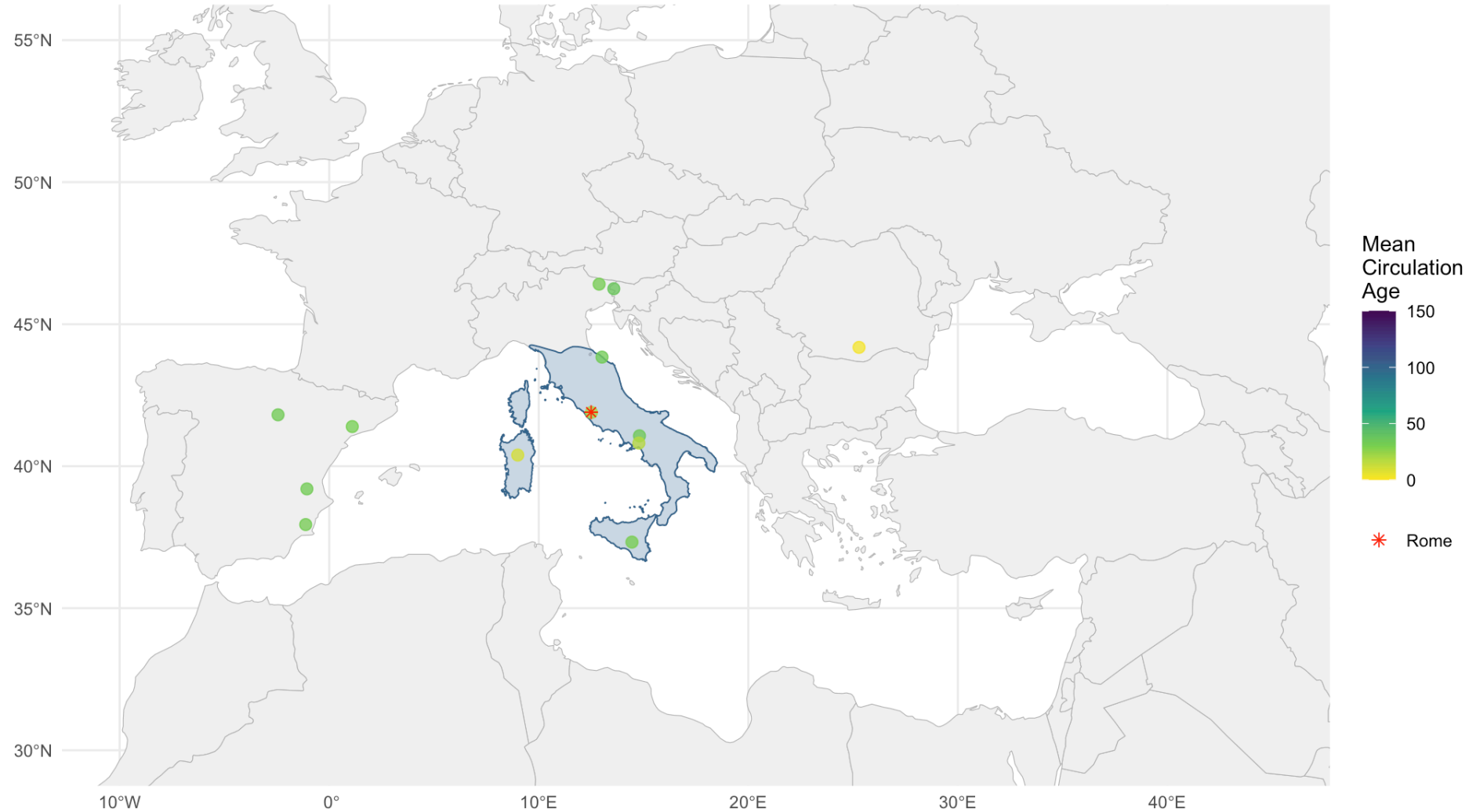
200 BCE 150 BCE 100 BCE 50 BCE 0 CE 50 CE

P1	200-146 BCE	Conquest	Mediterranean expansion; defeat of Macedon, Carthage, and Corinth
P2	146-100 BCE	Corruption	Post-conquest consolidation; Gracchan reforms; internal strain
P3	100-50 BCE	Crisis	Social War; Sullan civil wars; Mithridatic conflicts; escalating political violence
P4	50-30 BCE	Collapse	Terminal Republic; continuous civil war; competing monetary authorities
P5	30 BCE-50 CE	Comeback	Augustan settlement; renewed centralization and stability

Period - P1

P1 Conquest | Extent: 210 BCE

Mean circulation age: 26.4 years

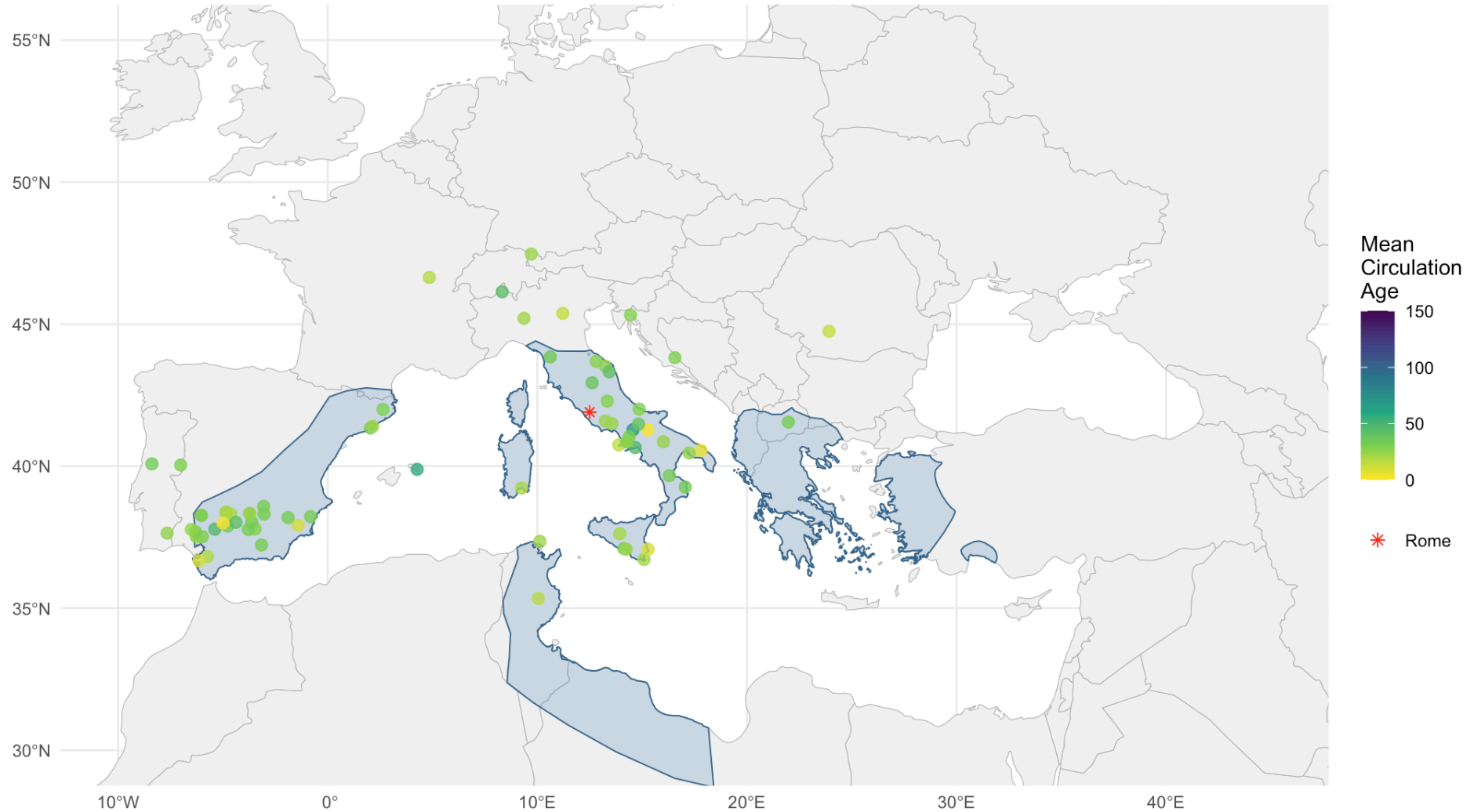


Mean Circulation Age is the mean of the hoard closing date minus the coin production date.

Period - P2

P2 Corruption | Extent: 129 BCE

Mean circulation age: 26.8 years

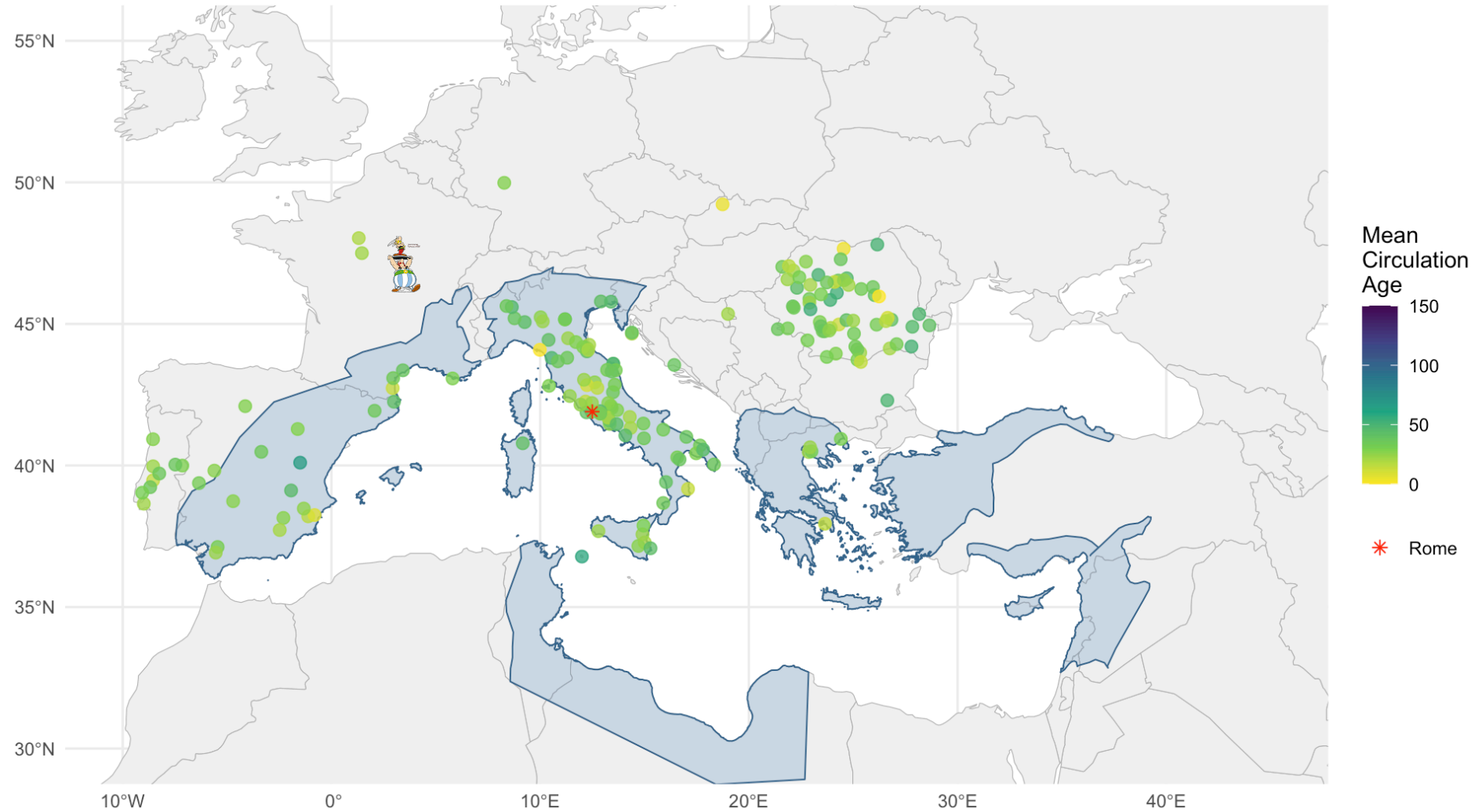


Mean Circulation Age is the mean of the hoard closing date minus the coin production date.

Period - P3

P3 Crisis | Extent: 58 BCE

Mean circulation age: 30.5 years

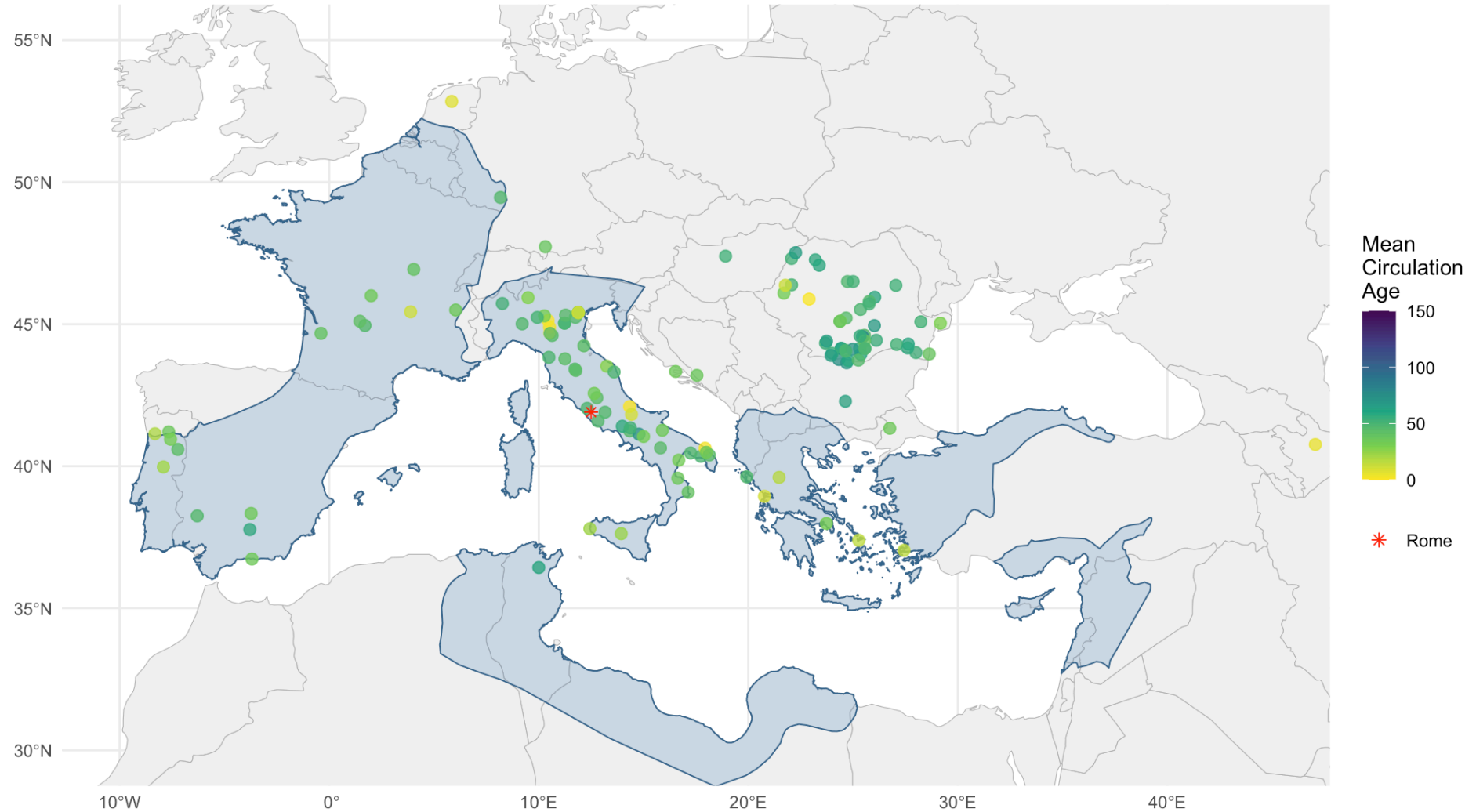


Mean Circulation Age is the mean of the hoard closing date minus the coin production date.

Period - P4

P4 Collapse | Extent: 44 BCE

Mean circulation age: 41.3 years

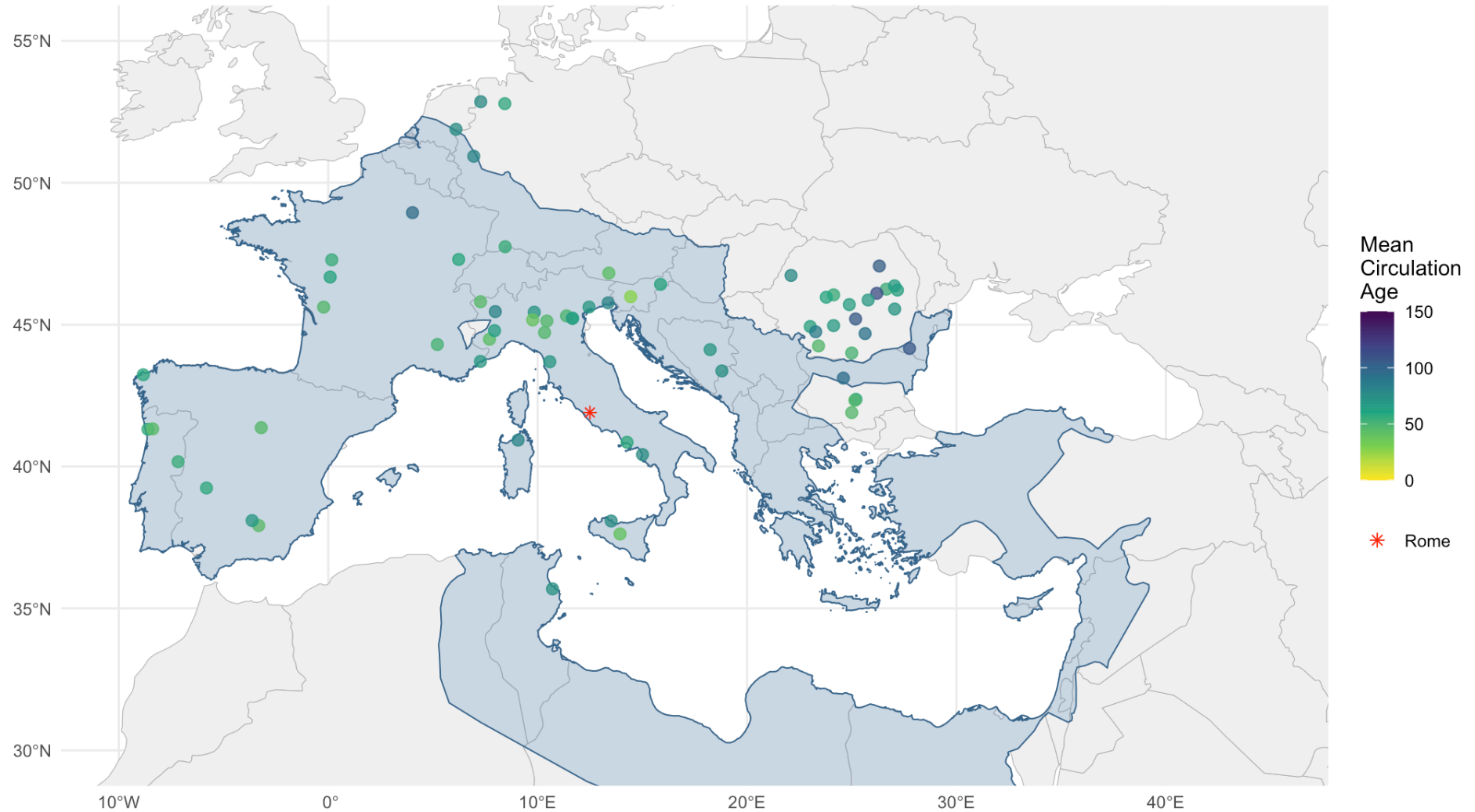


Mean Circulation Age is the mean of the hoard closing date minus the coin production date.

Period - P5

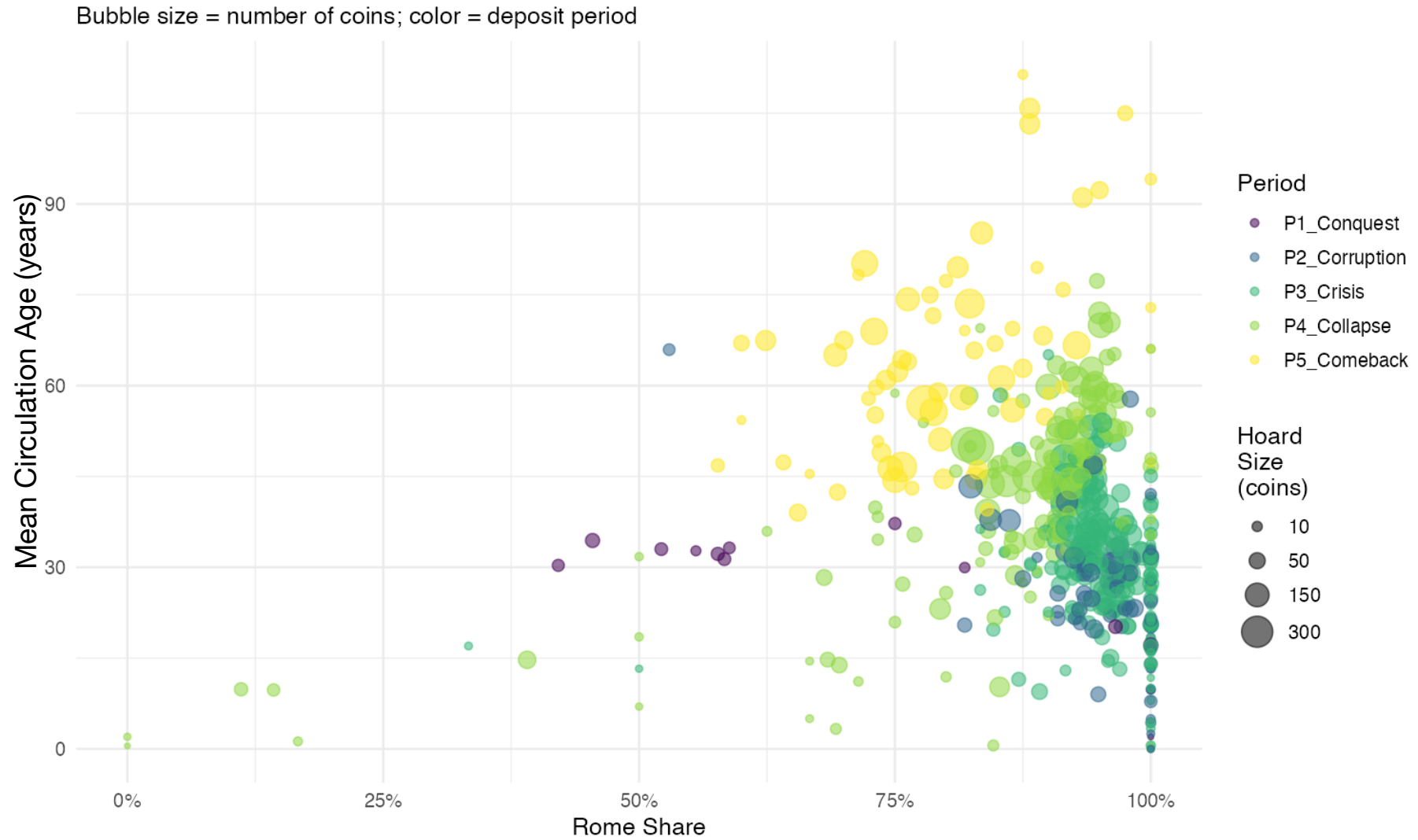
P5 Comeback | Extent: 14 CE

Mean circulation age: 63.8 years

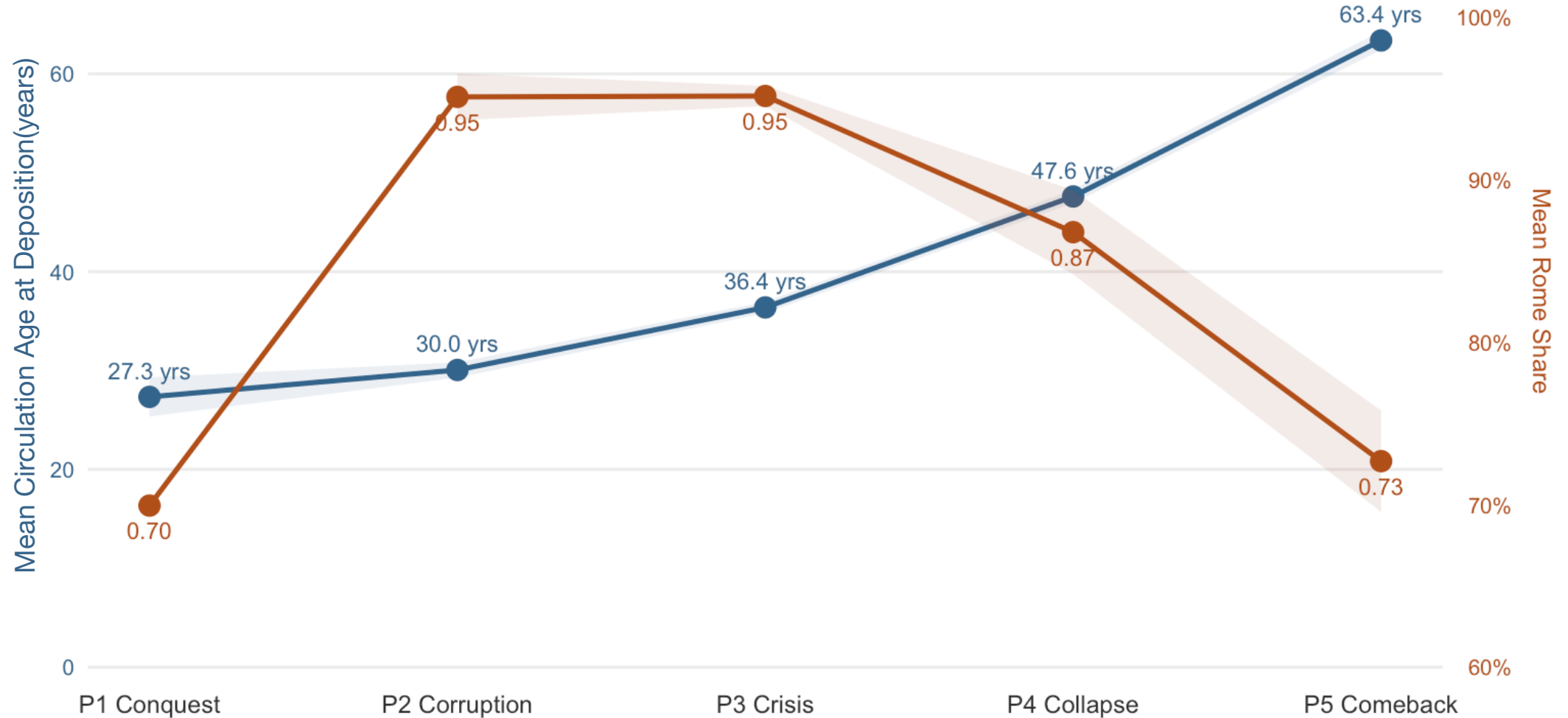


Mean Circulation Age is the mean of the hoard closing date minus the coin production date.

Exploratory Analysis

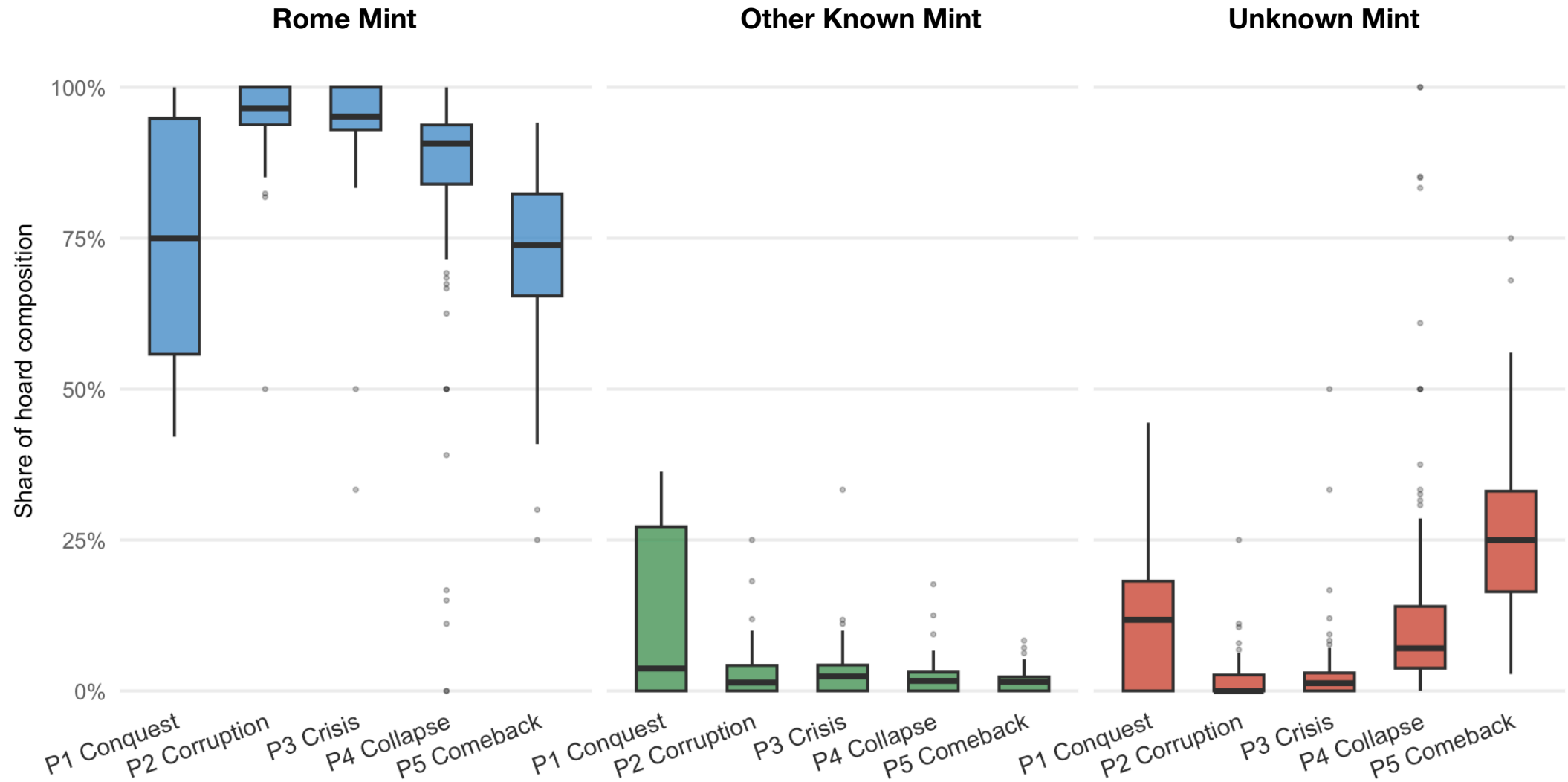


Exploratory Analysis



Source: CHRR (ANS). Coin age = TPQ minus production date midpoint. 54 negative values clamped to zero. Ribbon = 95% CI.

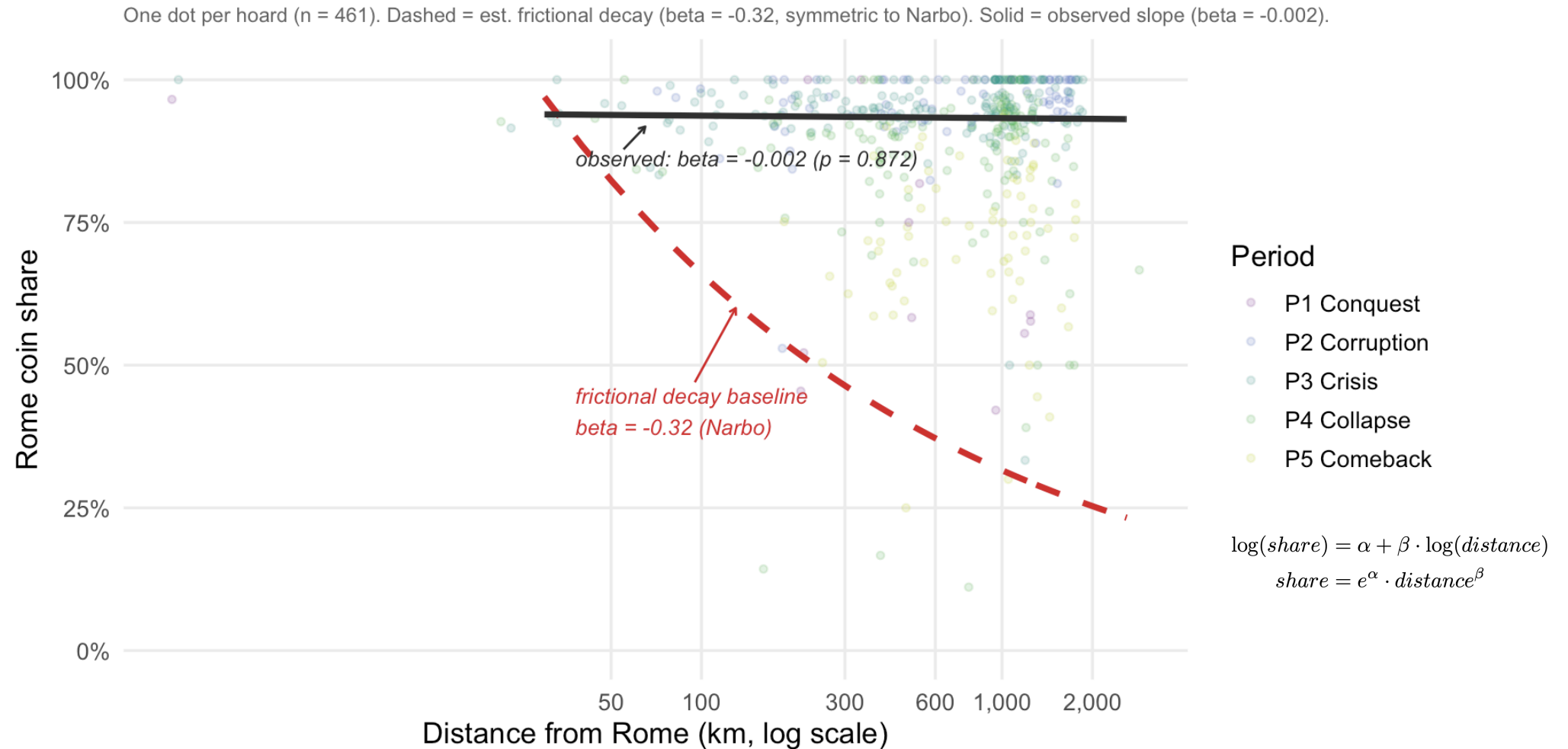
Mint Share



Empire-wide CHRR hoards with valid composition data. Unknown mint share includes uncertain_value attributions. Source: CHRR (ANS).

Distance Decay

Rome Share Does Not Decrease with Distance



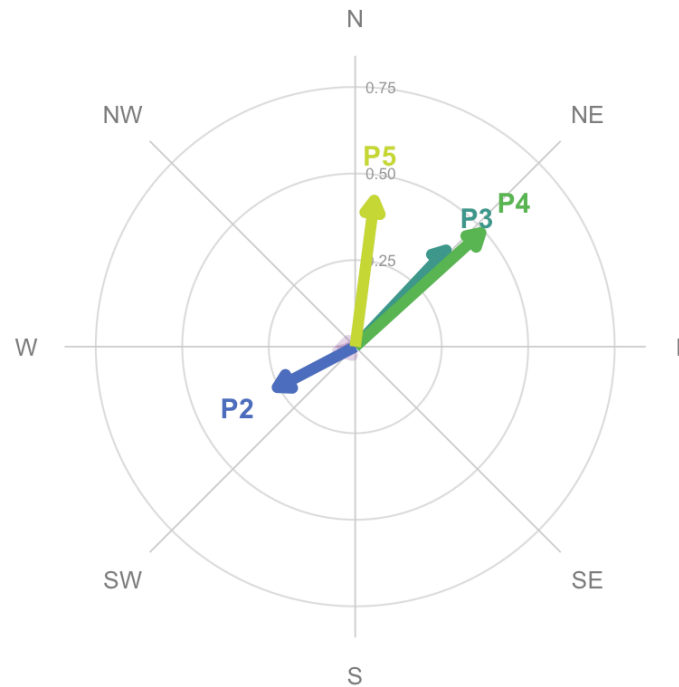
Hoard Direction from Rome

P5 - North

- The Cantabrian Wars, 29-19 BCE (W)
- The Conquest of the Alps, 15-14 BCE (N)
- The Pannonian Wars, 13-9 BCE (NE)
- The Dalmatian Wars, 6-9 CE (NE)
- The Germanic Campaigns, 12 BCE - 16 CE (N)

P2 - Southwest

- Post Punic Wars (SW)
- The Numantine War, 143-133 BCE (W)
- Transalpine Gaul, 125-121 BCE (SW)
- The Jugurthine War, 112-106 BCE (SW)
- The Slave Revolts, 135-132 BCE (S)



P3 - Northeast

- The Social War, 91-87 BCE (NE)
- First & Second Mithridatic Wars, 89-81 BCE (ENE)
- The Cimbrian War, 101 BCE (N/NE)
- Third Mithridatic War, 73-63 BCE (NE)

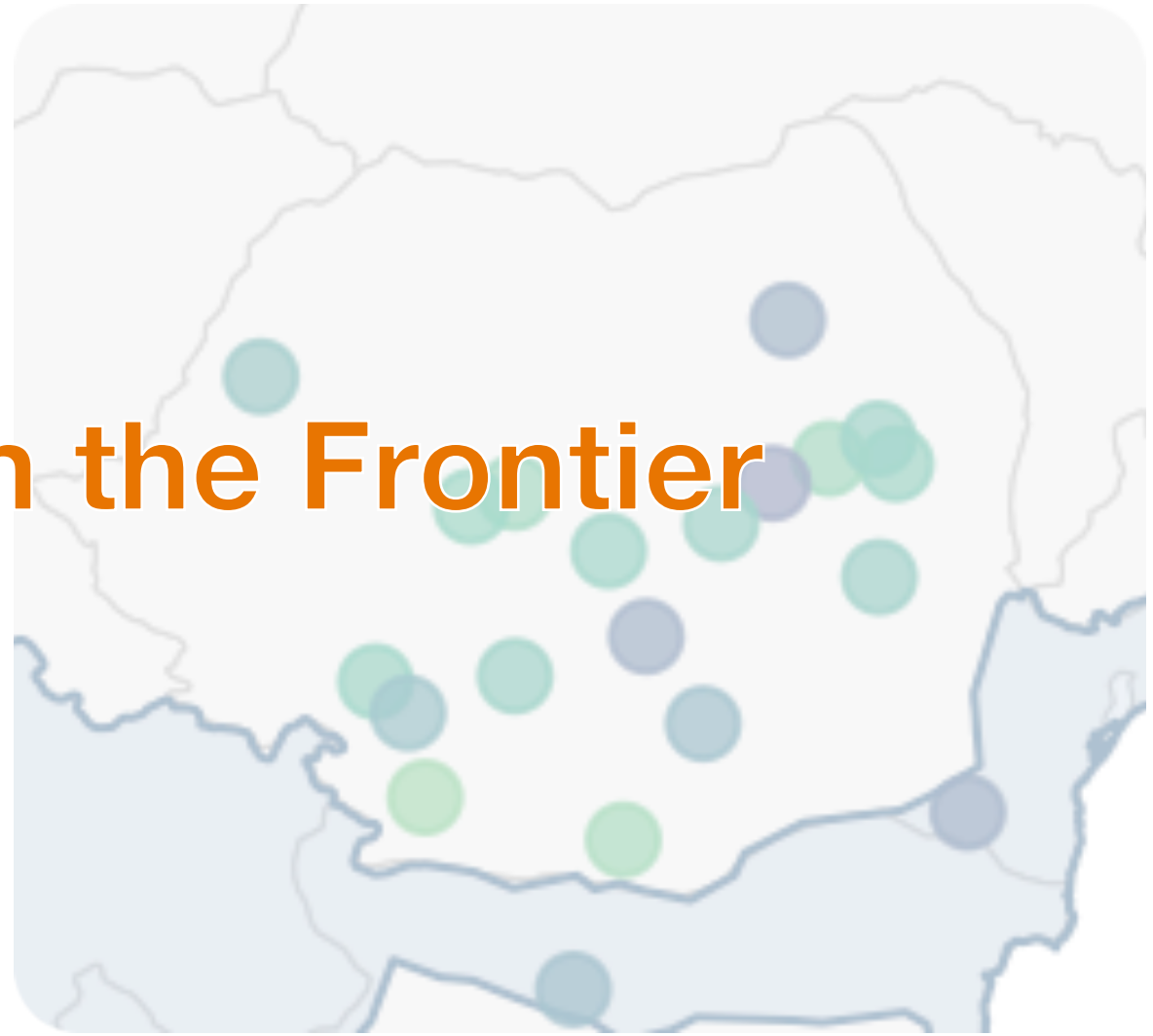
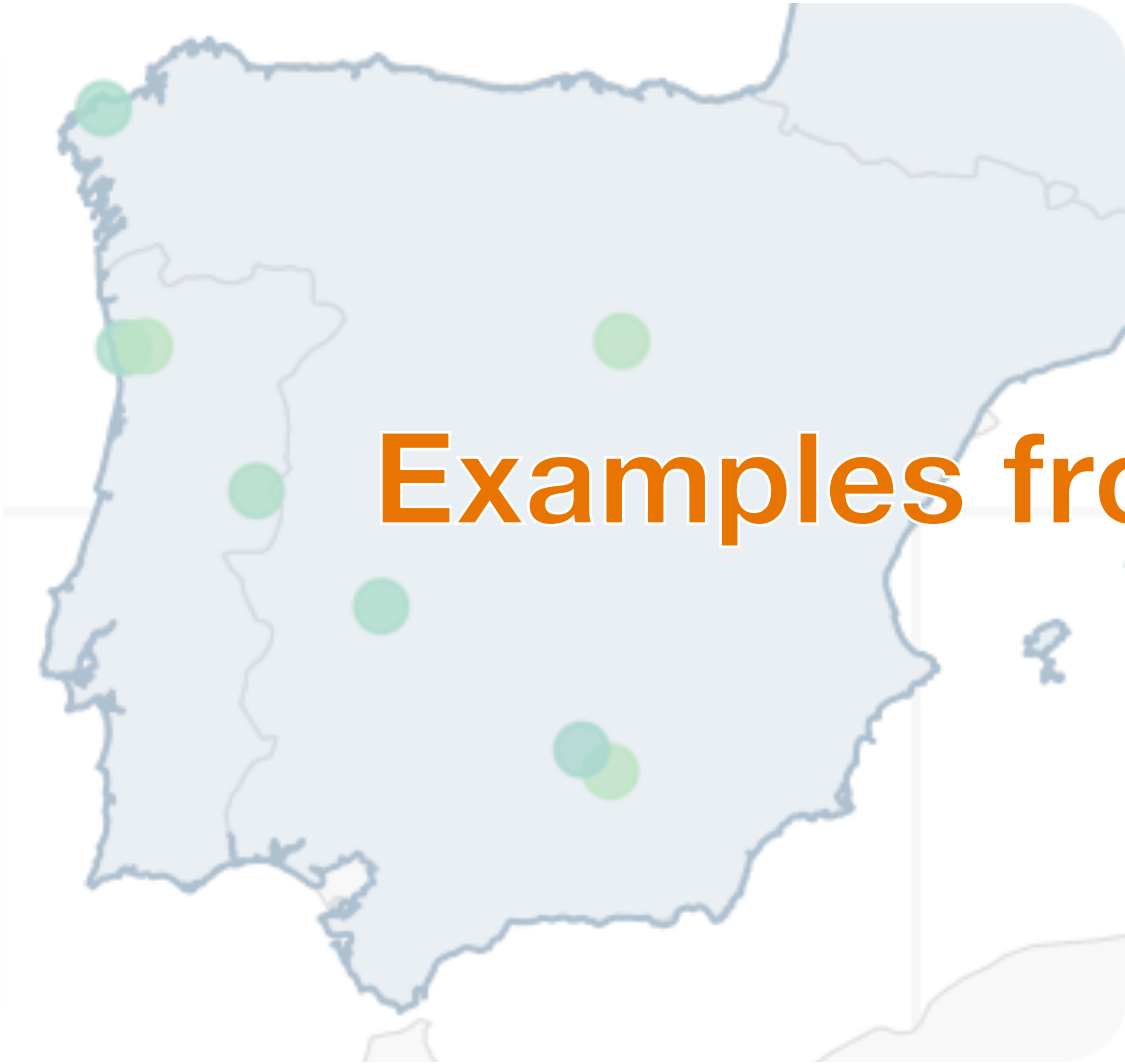
Period

- P1 Conquest
- P2 Corruption
- P3 Crisis
- P4 Collapse
- P5 Comeback

P4 - Northeast

- Caesar's Civil War, 49-45 BCE (NE)
- The Liberators' War, 44-42 BCE (NE)
- The Sicilian Revolt, 44-36 BC (SE)
- Parthian War, 36 BCE (E)
- The Final War of the Republic, 32-30 BCE (ENE)

Examples from the Frontier



Iberian Frontier

P1 Conquest | Extent: 210 BCE

Hoards: 13 | Weighted centroid trail through P1 Conquest

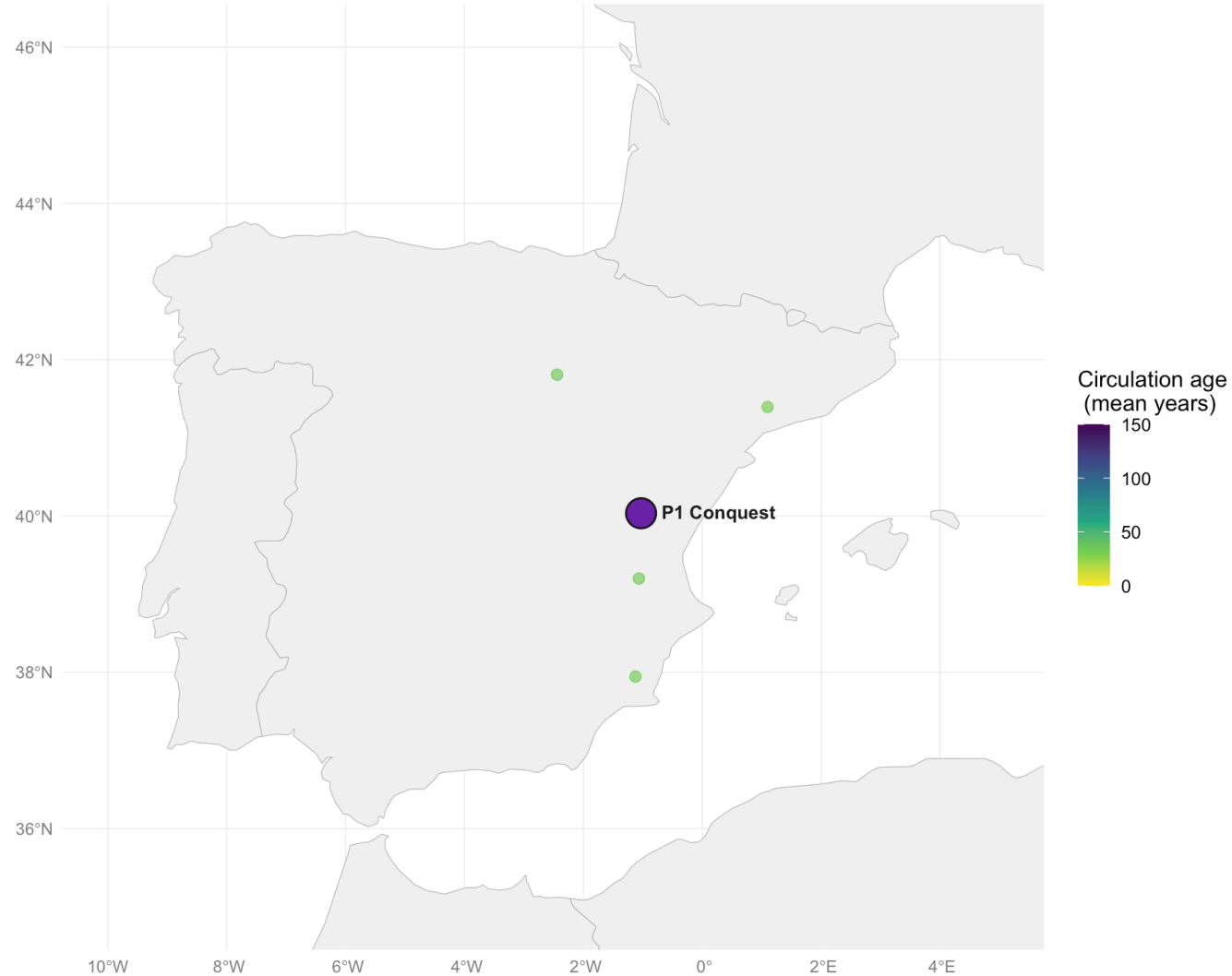
P1 Conquest

Exterior
Rome

Frontier
Rome

Interior
Rome

n=3
56%



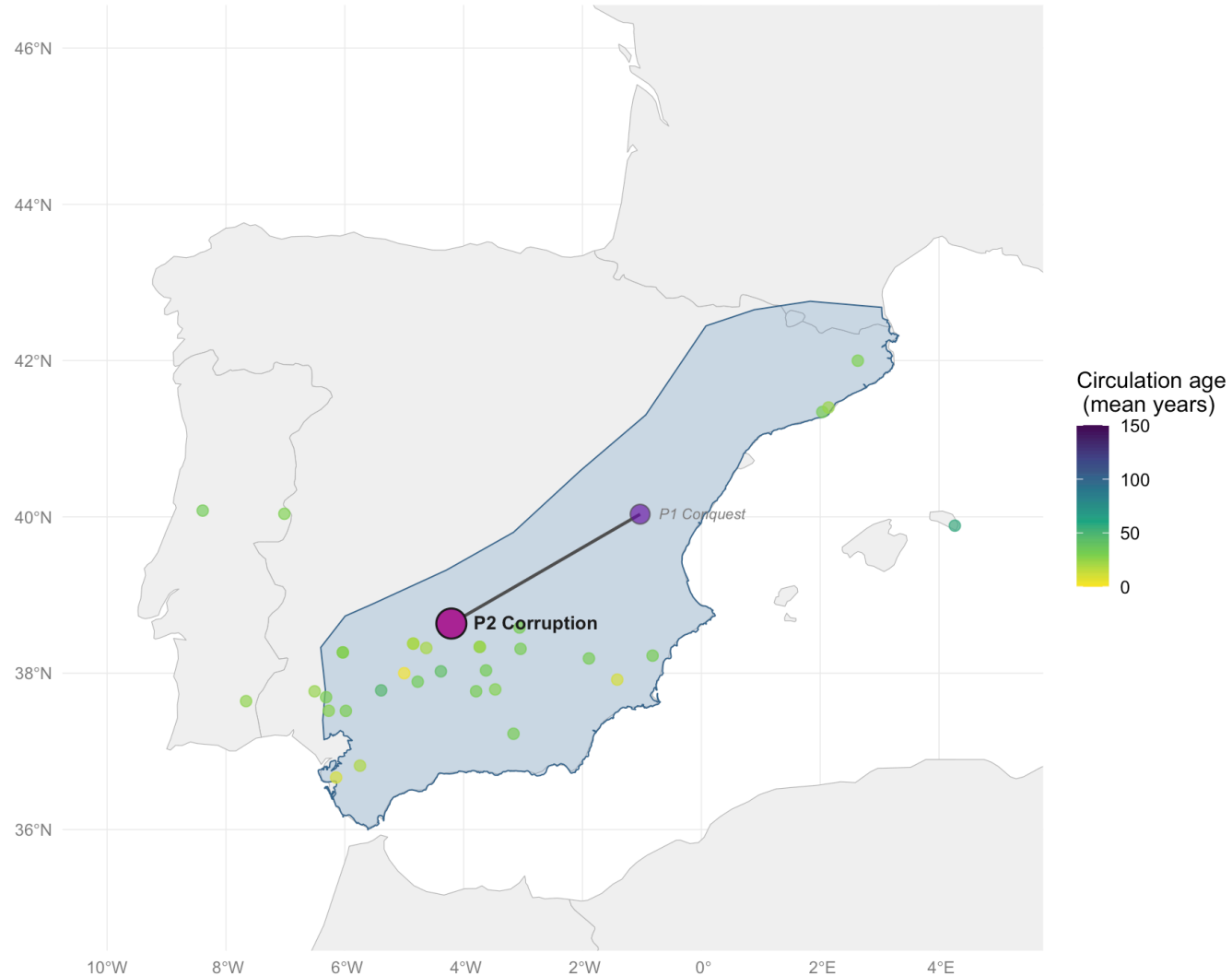
Centroids weighted by Rome share (Balearic hoards excluded). Arrows connect consecutive period centroids. Source: CHRR (ANS); Hollander shapefiles.

Iberian Frontier

P2 Corruption | Extent: 129 BCE

Hoards: 72 | Weighted centroid trail through P2 Corruption

P2 Corruption	
Exterior Rome	n=5 98%
Frontier Rome	n=12 97%
Interior Rome	n=4 96%



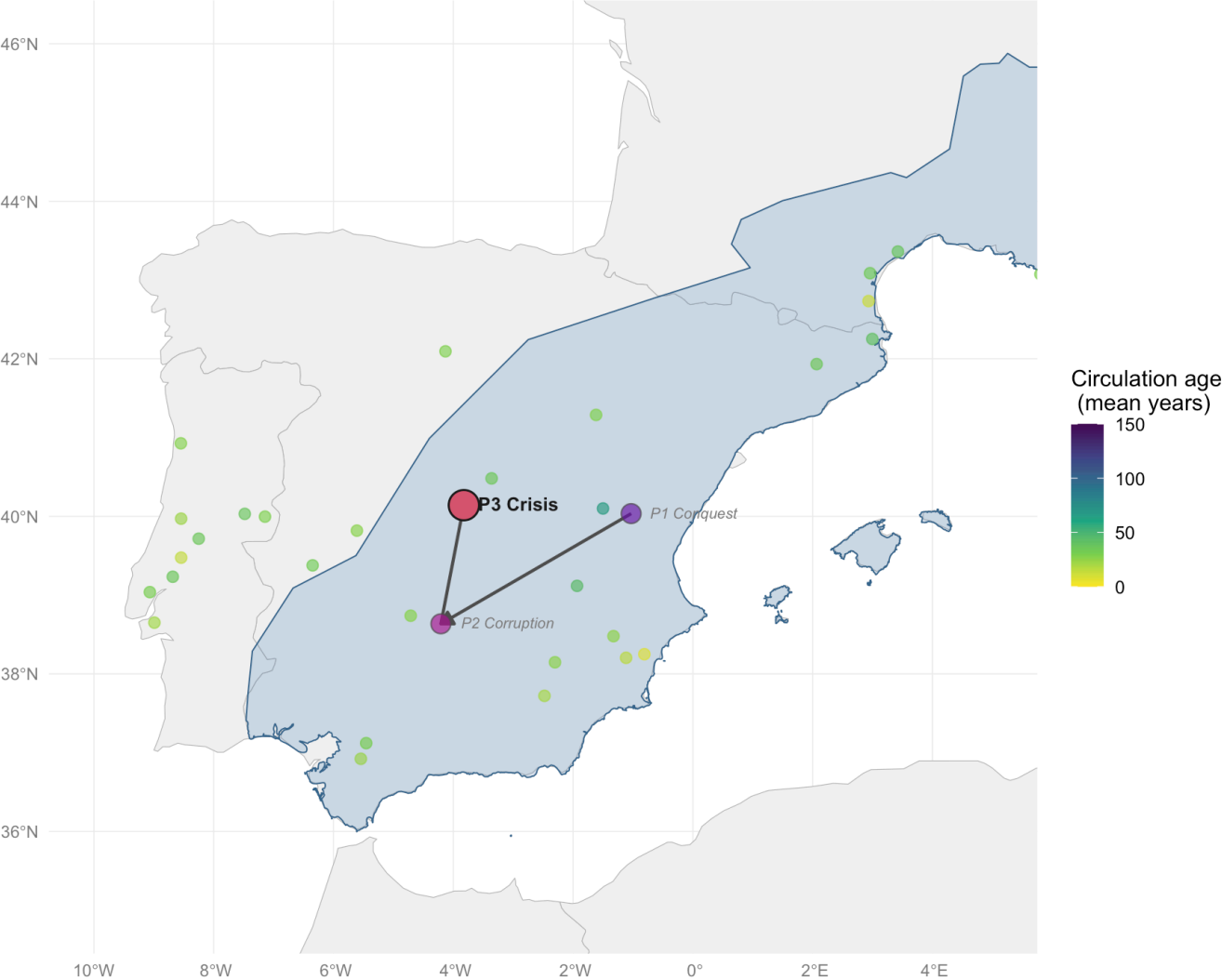
Centroids weighted by Rome share (Balearic hoards excluded). Arrows connect consecutive period centroids. Source: CHRR (ANS); Hollander shapefiles.

Iberian Frontier

P3 Crisis | Extent: 58 BCE

Hoards: 179 | Weighted centroid trail through P3 Crisis

P3 Crisis	
Exterior Rome	n=7 95%
Frontier Rome	n=12 92%
Interior Rome	n=4 96%



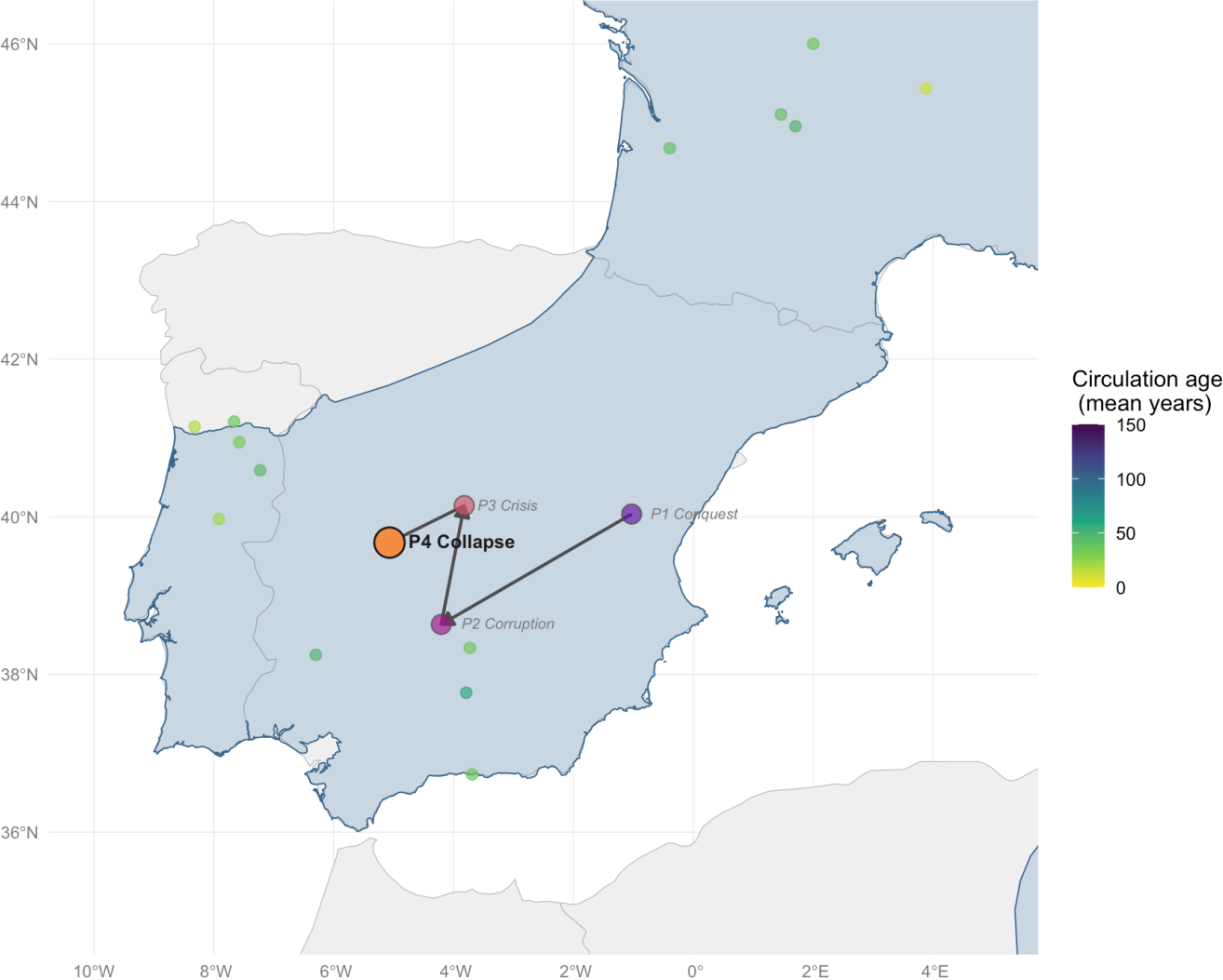
Centroids weighted by Rome share (Balearic hoards excluded). Arrows connect consecutive period centroids. Source: CHRR (ANS); Hollander shapefiles.

Iberian Frontier

P4 Collapse | Extent: 44 BCE

Hoards: 131 | Weighted centroid trail through P4 Collapse

P4 Collapse	
Exterior Rome	
Frontier Rome	n=4 86%
Interior Rome	n=4 82%



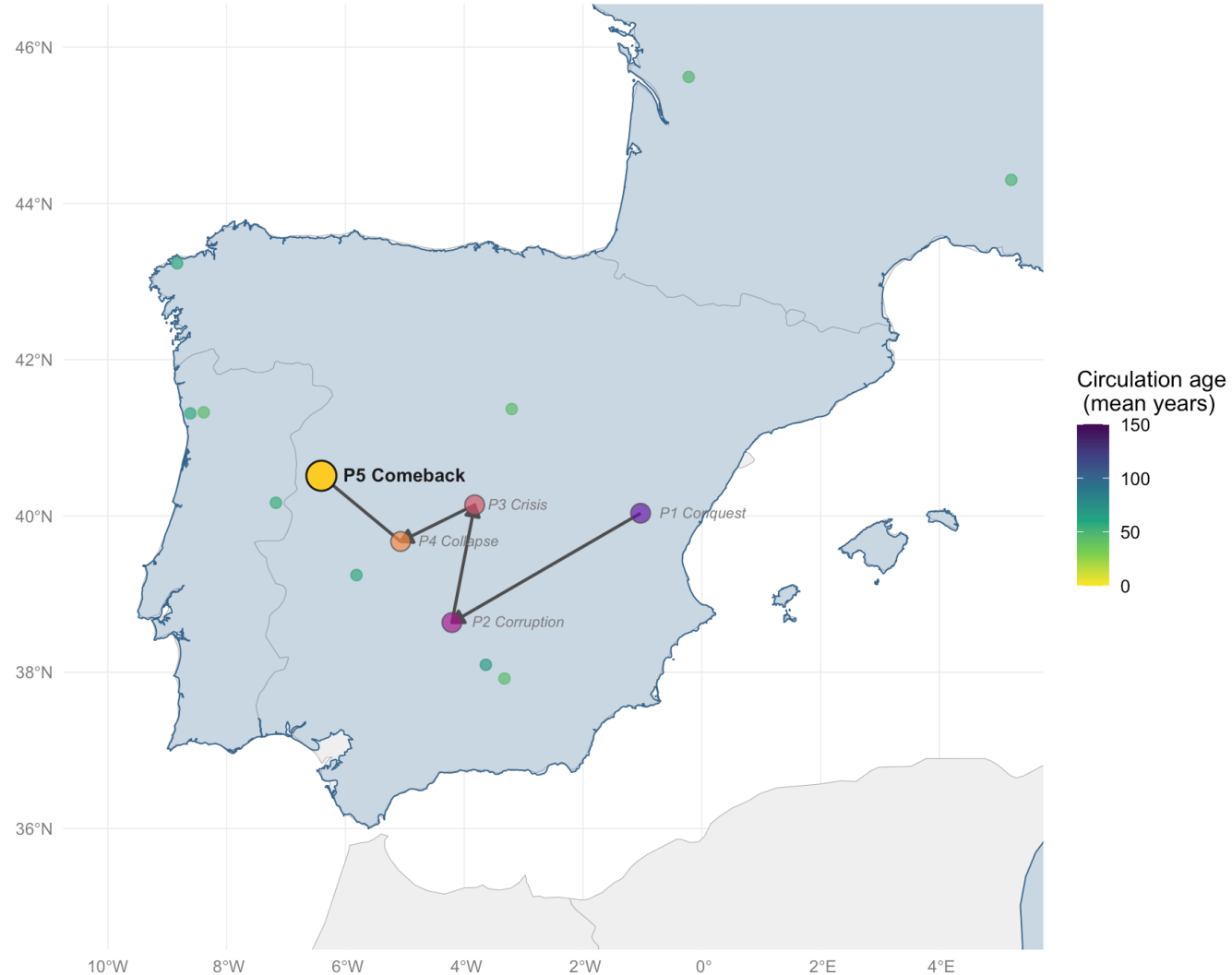
Centroids weighted by Rome share (Balearic hoards excluded). Arrows connect consecutive period centroids. Source: CHRR (ANS); Hollander shapefiles.

Iberian Frontier

P5 Comeback | Extent: 14 CE

Hoards: 68 | Weighted centroid trail through P5 Comeback

P5 Comeback	
Exterior Rome	
Frontier Rome	n=3 75%
Interior Rome	n=5 57%



Results

- Nothing showed no systematic trend
- Easting showed a weak negative trend at $p = 0.06$

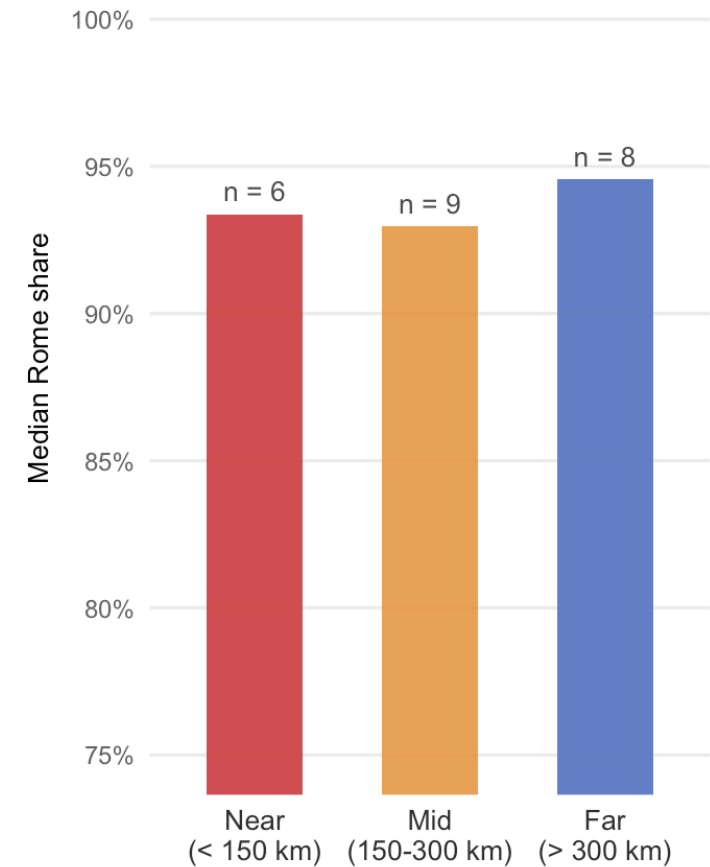
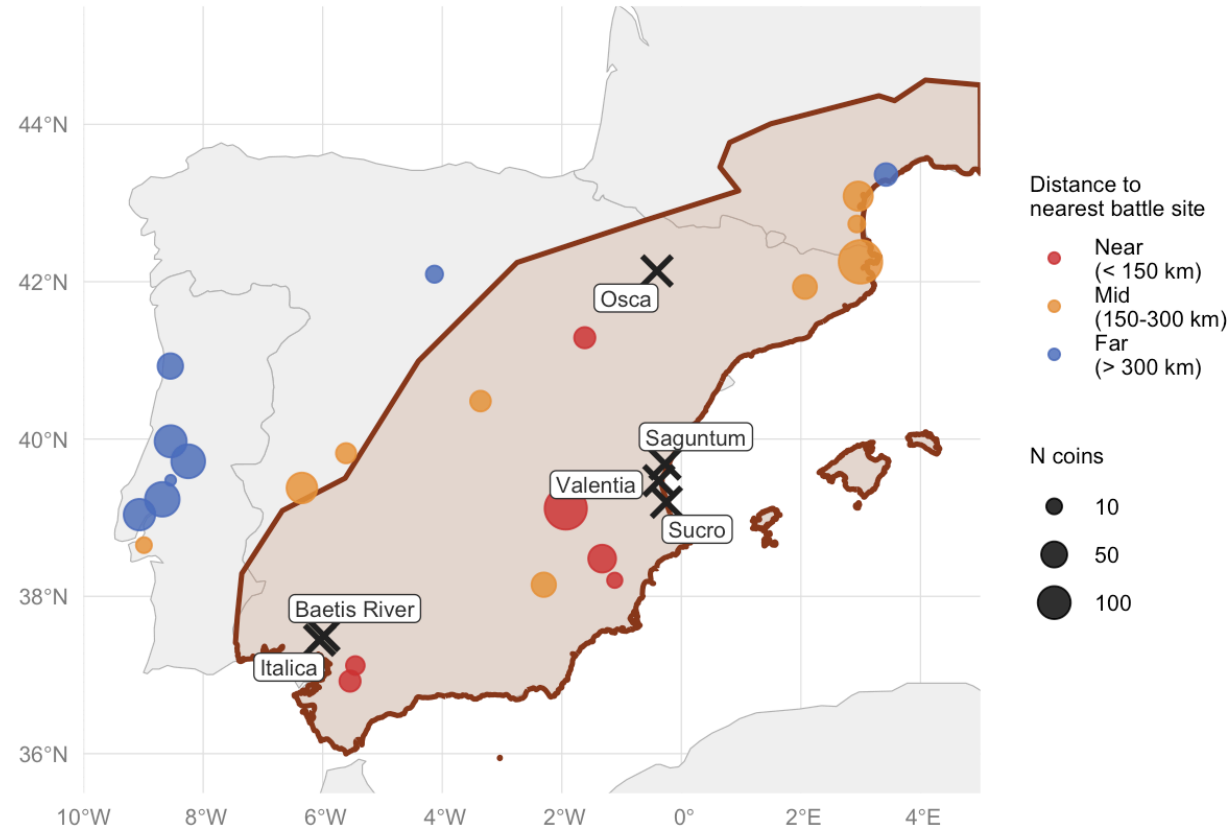
Centroids weighted by Rome share (Balearic hoards excluded). Arrows connect consecutive period centroids. Source: CHRR (ANS); Hollander shapefiles.

Iberia - Sertorian War

The Sertorian War left no detectable imprint on the coin record

P3 Crisis hoards (n = 23) vs Sertorian War battle sites (80-72 BCE)

Rome share does not vary with proximity to battle site

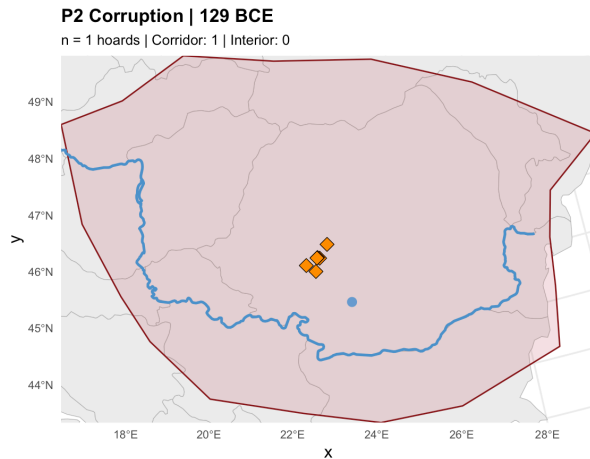


Battle sites geocoded from Pleiades. Error bars = interquartile range. Source: CHRR (ANS); Pleiades.

Frontier - Dacia

P1 Conquest

- Hoard Density in Dacia was non-existent
- Little commerce between Italy and Dacia

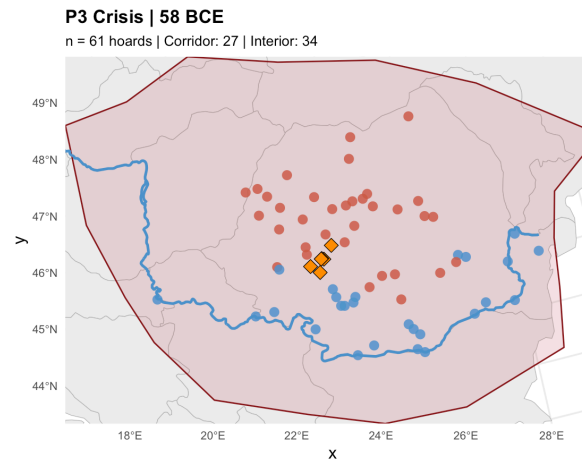


P2 Corruption

- Delos in the Aegean became a major trade center
- No significant hoarding

P3 Crisis

- Mithridates destroyed Delos in 88 BCE disrupting the Aegean commercial infrastructure
- Pompey suppressed Cilician piracy in 67 BCE
- Greek Black Sea colonies began connecting Roman coin to the Dacian interior
- Burebista united a fragmented Dacia



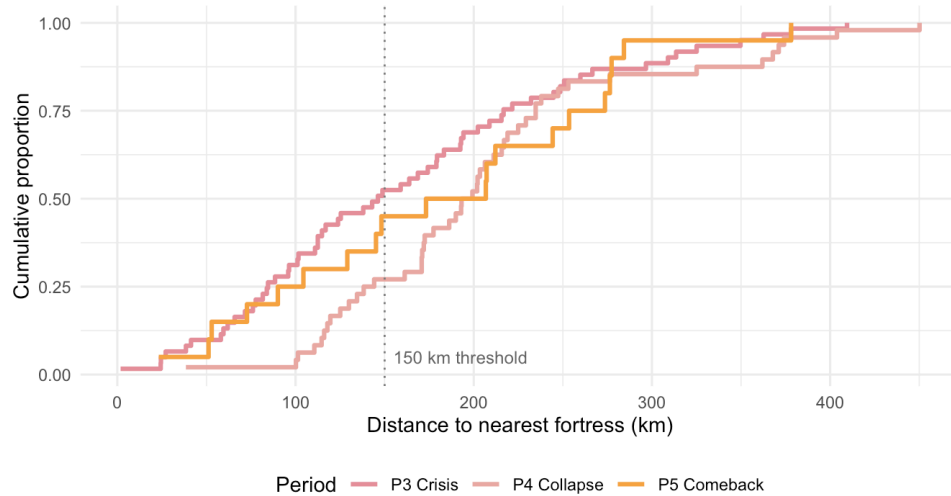
P5 Comeback

- Increased trade on the Danubian frontier
- Fragmentation of Dacia

P4 Collapse

- Transylvanian gold and silver attracted Rome
- Hoard accumulation in Romania accelerated
- Burebista killed

Dacia: Confederation & Collapse



Median Fortress Distance

- P3 distance: 146 km (n = 61)
- P4 distance: 196 km (n = 48)

Wilcoxon rank sum: $W = 1020$, $p = 0.007$

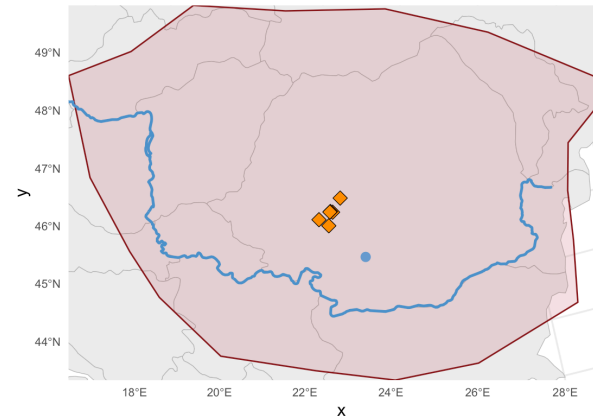
P4 to P5: no further change (~190 km, $p = 0.554$)

Danube corridor share at 100 km: P3 = 44%, P4 = 60%

Knox test: Not significant across all threshold combinations which means there was sustained, not episodic deposition. Fail to reject that space and time are independent.

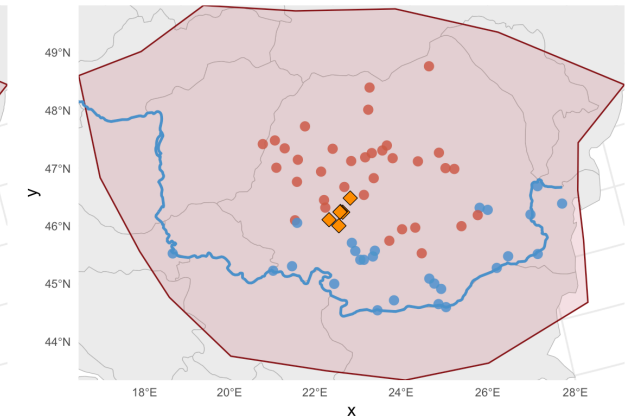
P2 Corruption | 129 BCE

n = 1 hoards | Corridor: 1 | Interior: 0



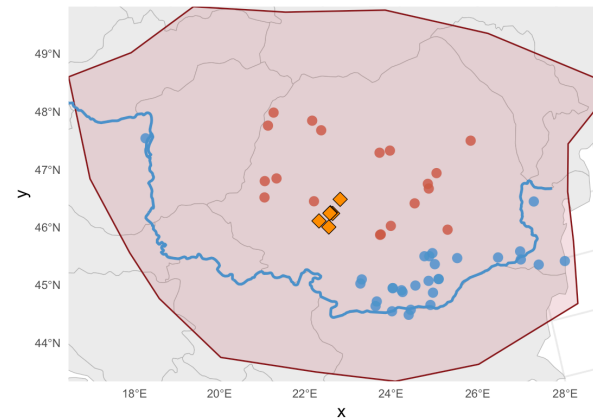
P3 Crisis | 58 BCE

n = 61 hoards | Corridor: 27 | Interior: 34



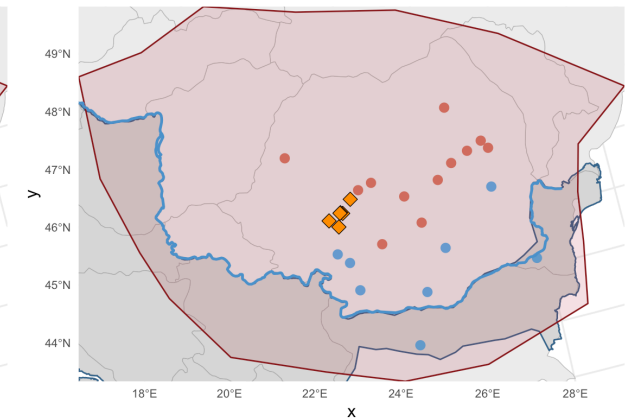
P4 Collapse | 44 BCE

n = 48 hoards | Corridor: 29 | Interior: 19



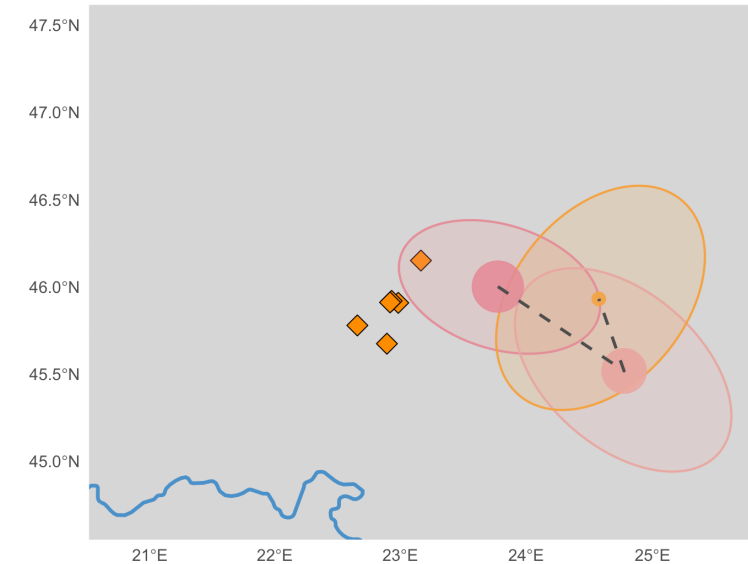
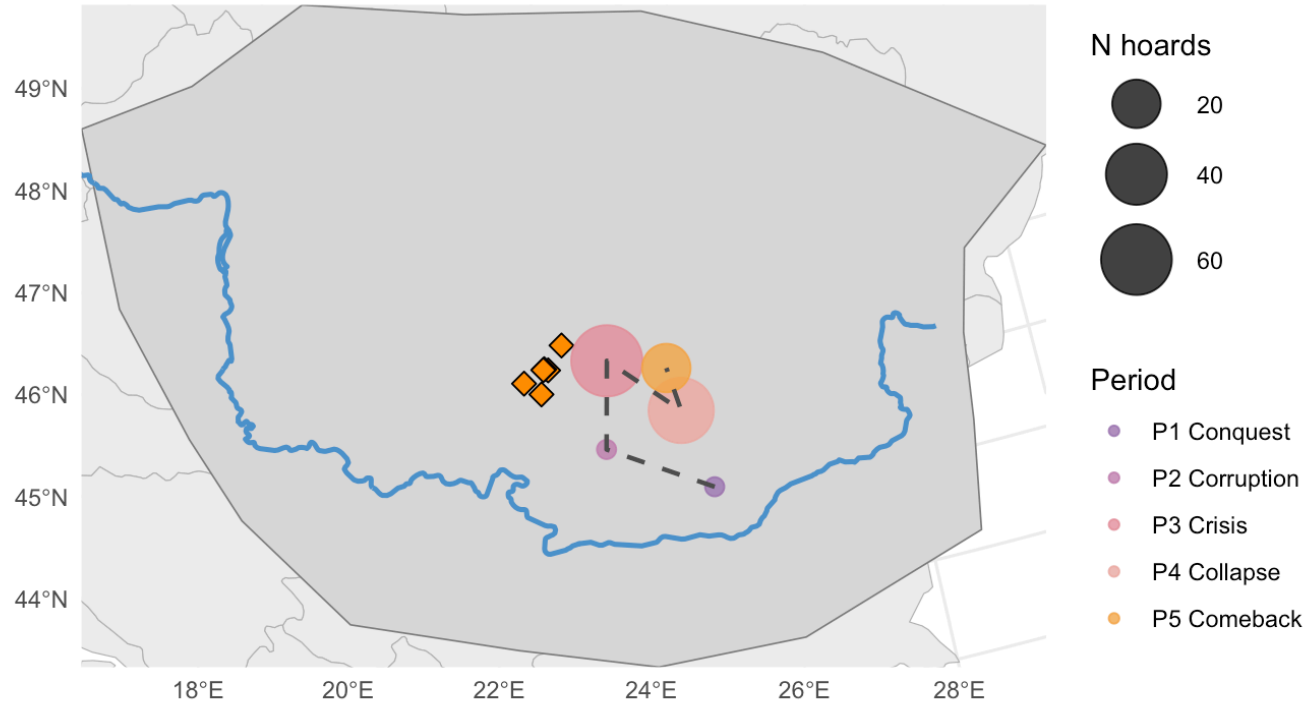
P5 Comeback | 14 CE

n = 20 hoards | Corridor: 8 | Interior: 12



● Corridor ● Interior □ Fortress ■ Roman Empire ■ Dacian Confederation

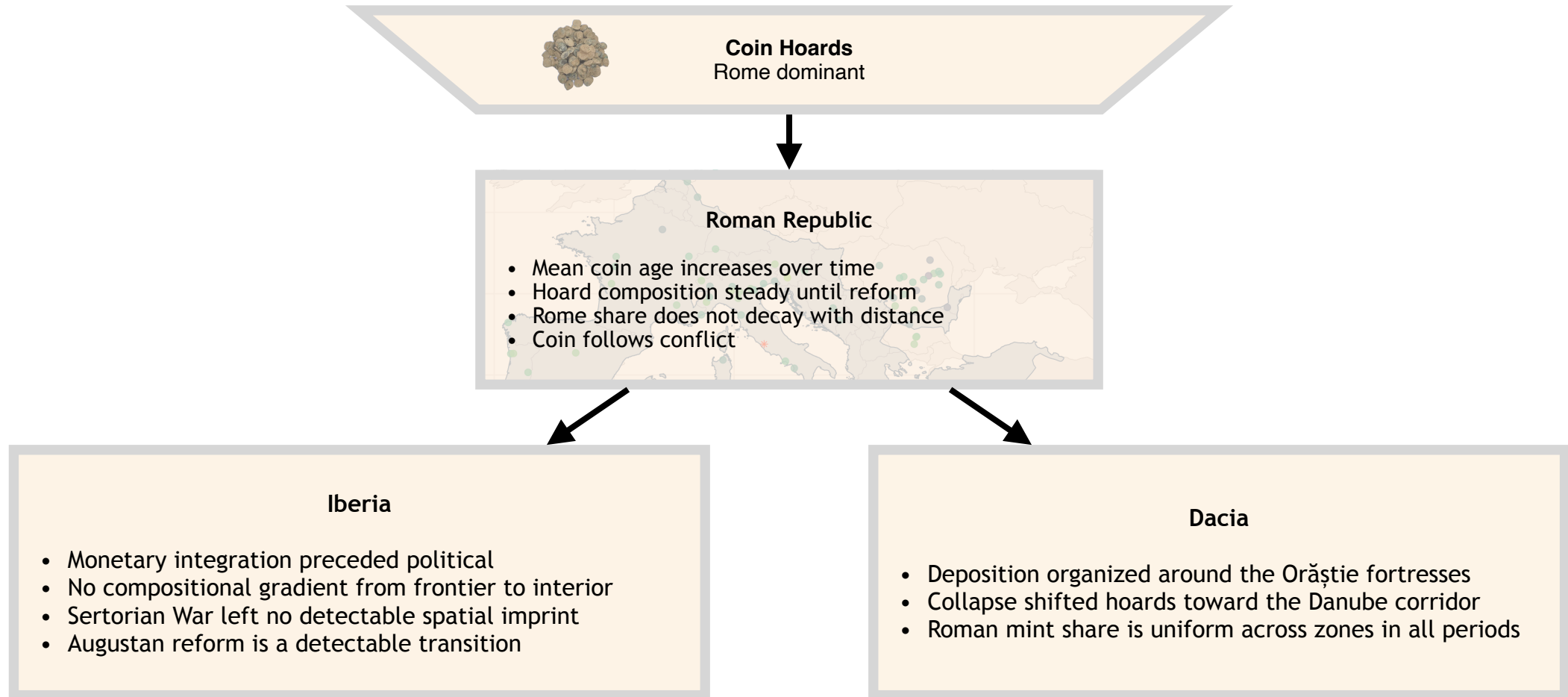
Dacia: Centroid Drift



Comparison	Shift (km)	<i>p</i>
P3 Crisis to P4 Collapse	97.8	0.06
P4 Collapse to P5 Comeback	49.2	0.58

Permutation test for centroid shift between consecutive periods. The observed shift (km) is compared to 999 random reassignments of hoards to periods. The p-value measures how often a random reassignment produces a shift as large as the observed one.

Summary



Sample of Analyses Performed

Composition & Interpretation						
ID	Methodology	Key Variables	Finding	Result	Periods	Region
m0.5	Evidence synthesis / decision framework	Period segmentation validation	Period boundaries correspond to structural breaks; justifies period-specific analysis over pooled models	Framework	All periods	Republic-wide
m0.8	Evidence synthesis / decision framework	Model-path decision	Global single-process models rejected; period-specific compositional and regional analyses warranted	Framework	All periods	Republic-wide
H11.2	OLS regression (ORBIS vs Euclidean)	rome_share vs ORBIS road distance	Road-only outperforms full network and Euclidean for continental mints; sea routes not primary diffusion channel	Significant	All periods	Republic-wide
H11.3	Anisotropy (Rayleigh test)	Directional distribution of provincial-mint hoards from Rome	Strong flow bearing NNW; directional concentration significant	Significant	All periods	Republic-wide
H14.2	Wilcoxon rank-sum (fortress distance P3 vs P4)	Fortress distance shift	Significant dispersal from fortress zone between P3 and P4	Significant	P3-P4	Dacia
H14.3	Danube corridor share by buffer width	Proportion of hoards within Danube buffer (50/100/150 km)	Corridor share increases from 55% to 70% at 150 km buffer (P3 to P4); robust across all three thresholds	Significant	P3-P4	Dacia
H15.1	Rome share by spatial zone	rome_share across corridor vs interior zones	NS in all testable periods; composition uniform regardless of spatial zone	Null (informative)	By period	Dacia
H16.2	Wilcoxon rank-sum (Rome share by zone)	rome_share across land-only frontier / interior zones	No Rome share gradient; null result stronger than H13 because it survives stricter zone definition	Null (informative)	All periods	Iberia
H16.3	P4 to P5 compositional shift	rome_share and unknown_share transition	Rome share decline; unknown share rise	Significant	P4-P5	Iberia
H17.1	OLS regression (pooled)	rome_share vs Euclidean distance from Rome	No distance decay: Rome share does not decline with distance	Null (informative)	Pooled	Republic-wide
H17.6	P1 Conquest contrast (H17 vs H11)	Directional concentration: Rome-mint vs provincial-mint hoards	H11 provincial coins NNW vs H17 Rome coins isotropic	Significant	P1	Republic-wide
H18.4	Global Moran I (rerun)	Distance to Danube (km)	Bimodal structure attenuates under Augustus	Confirmed	P3-P5	Dacia
D2.5	Deposit vs circulation signal separation	rome_share spatial autocorrelation vs N_eff spatial autocorrelation	Empirical basis for claim that composition and location carry independent signals; compositional nulls reflect uniform input not absence of signal	Framework	Pooled	Republic-wide



Contributions and Lessons

- **Empirical Signal Separation:** Separated the historical signals of coin circulation from hoard deposition. This provided a statistical test for the theoretical distinction identified by Buttrey (1999) using formal spatial methods.
- **Diagnostic Workflow:** Established a process to assess whether sparse or fragmentary archaeological data contains sufficient signal to warrant spatial inference before committing to complex models.
- **Transferable Framework:** Developed a reusable methodology for evaluating incomplete datasets, ensuring that analysis is focused only on areas where spatial signatures are statistically robust.
- **Selective Signal Recovery:** Demonstrated that spatial analysis can pinpoint which historical mechanisms left a durable imprint on the archaeological record and which did not.
- **Informative Nulls:** Demonstrated that the absence of a spatial pattern is a substantive finding. Nulls on distance decay, frontier gradient, and Sertorian imprint eliminated specific mechanisms and redirected analysis toward where spatial signal actually resided.
- **Non-Stationarity as Finding:** The failure of a global model is itself a historical result rather than a problem. The instability of empire-wide parameters opened the regional analyses that revealed Iberia and Dacia as structurally distinct spatial processes.
- **Period Stratification:** Temporal stratification keyed to historical transitions recovers structure that pooled analysis obscures. Significant empire-wide spatial clustering appeared only in the Collapse period and would have been invisible under an undifferentiated analysis of the full dataset.
- **Comparison Before Test:** Specifying the correct comparison matters. Mathematically equivalent variables and alternative distance specifications produced divergent results, and the interpretive work lay in choosing the question rather than the procedure.

SATIS DICTUM EST